Lakshya Gupta

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Education

Vellore Institute of Technology — Vellore, Tamil Nadu
BTech in Computer Science Engineering, specializing in Cybersecurity — *June 2025 - June 2029*

Experience

McKinsey Forward — September 2026 - Present
Forward Learner

- Engaged in McKinsey's Forward program, building structured problem-solving, hypothesis-driven thinking, and client-communication skills through guided modules and practical exercises

Projects

GridSense — AI-Powered Power Grid Risk Prediction & Prevention — August - October 2026
Python | PyTorch | GAT | Scikit-learn | Pandas | FastAPI | React | PostgreSQL source

- Developed a GAT-based power-grid risk prediction system analyzing ~22K assets and ~23K transmission lines to identify potential cascading failures, achieving ~98% accuracy in identifying high-risk grid conditions during model evaluation.
- Built a Predict → Propagate → Pinpoint → Prevent pipeline combining failure propagation, root-cause analysis, physics-based validation, and what-if simulations.
- Implemented early-warning, anomaly detection, RUL estimation, and asset-level risk scoring to support proactive grid failure prevention and maintenance.

AI DevOps Agent - Kubernetes Auto-Healing — September 2026
Kubernetes, Python, LangGraph, Ollama, Prometheus, Grafana, Docker source

- Automated Kubernetes failure detection and diagnosis, analyzing logs, metrics, and cluster events to reduce manual troubleshooting effort by ~60%
- Implemented AI-driven remediation workflows using LangGraph and Kubernetes APIs, reducing recovery time by ~50% through automated root-cause analysis and corrective actions
- Architected a closed-loop auto-healing system that monitors, remediates, and verifies workload recovery, achieving ~80% faster incident response in simulated failure scenarios

BhoomiSeva - National Digital Land Governance Platform — InnoHack 2.0 — August 2026
Flask, SQLite, Python, D3.js | Team DEBUGGERS source

- Built a full-stack geospatial registry platform with parcel records, an interactive map explorer, and grievance filing/tracking
- Implemented a server-side map link resolver, cascading location endpoints, and a robust CSV ingestion pipeline with dynamic schema detection
- Developed an sklearn-based price forecasting module for real per-parcel land value regressions, with a dark-themed D3.js map frontend

Network Intrusion Detection System (NIDS) — October 2026
Python, Scapy, FastAPI, Isolation Forest, React, SQLite source

- Developed a network security system using Python, Scapy, and FastAPI to analyze PCAP files and monitor suspicious network traffic.
- Implemented detection for port scanning, brute-force attacks, and network anomalies using rule-based techniques and Isolation Forest.
- Built an interactive React dashboard with threat alerts, risk levels, and traffic statistics, using SQLite to store analysis results.

Certifications & Achievements

Hackathon Achievements

- **InnoHack 2.0** — **2nd Runner-Up**
- **WeHack 5.0** — Top 15 Finalist Teams
- **VinHack** — Top 15 Finalist Teams
- **Hack Battle** — Top 20 Teams

AI Tools & ChatGPT Workshop - be10X — June 28, 2026

- Completed a certified workshop covering practical use of AI tools: creating presentations in under 5 minutes, analyzing data in under 30 minutes, and coding/debugging with AI assistance in under 10 minutes

Technical Skills

Languages: Java, Python, C++, C, HTML, CSS
AI / DevOps Tools: LangGraph, Ollama, Kubernetes, Docker, Prometheus, Grafana
Core Concepts: OOP, Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Cybersecurity Fundamentals
Version Control: Git, GitHub
IDEs / Editors: VS Code, IntelliJ IDEA, Eclipse
Dev Tools & Platforms: Postman, MySQL, PostgreSQL, Linux/CLI
Productivity: Project Management, AI-Assisted Development Tools (ChatGPT, Claude)